

Wasserstein Polyak–Łojasiewicz Inequalities, Linear Convergence, and Functional Inequality Examples

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Abstract

This note records a useful dictionary: a Wasserstein Polyak–Łojasiewicz inequality is exactly an entropy–entropy production inequality for the chosen Wasserstein energy. Under the usual energy dissipation identity for the Wasserstein gradient flow, it gives exponential decay of the energy gap and, by a tail-length argument, exponential convergence in Wasserstein distance to a minimizer selected by the flow. The exponent is 2κ for the energy gap and κ for the W_2 distance itself. Positive Wasserstein geodesic convexity implies the Wasserstein–PL inequality, but the converse is false. A concrete counterexample is given by the potential energy generated by the nonconvex one-dimensional potential

$$V(x) = x^2 + 2 \sin^2 x.$$

The last section lists several classical and nonclassical functional inequalities that are Wasserstein–PL inequalities for particular energies, explains how the distance estimate specializes to Talagrand-type transportation inequalities in entropy examples, and clarifies the relation with Poincare-type inequalities. In particular, logarithmic Sobolev is the nonlinear entropy/Fisher-information analogue of the Poincare spectral-gap inequality; it implies Poincare by linearization, but the converse is false.

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1 Setting and notation

Let M be either $\mathbb{R}^d$ or a smooth complete Riemannian manifold, and let $\mathcal{P}_2(M)$ denote the probability measures with finite second moment, endowed with the 2-Wasserstein distance W_2 . For a functional

$$\mathcal{F} : \mathcal{P}_2(M) \rightarrow (-\infty, +\infty]$$

we write

$$\mathcal{F}_\star := \inf_{\rho \in \mathcal{P}_2(M)} \mathcal{F}(\rho), \quad \text{Argmin } \mathcal{F} := \{\rho : \mathcal{F}(\rho) = \mathcal{F}_\star\}.$$

The metric slope of $\mathcal{F}$ at ρ is

$$|\partial\mathcal{F}|(\rho) := \limsup_{\eta \rightarrow \rho} \frac{(\mathcal{F}(\rho) - \mathcal{F}(\eta))_+}{W_2(\rho, \eta)}. \quad (1)$$

When ρ has a density and $\mathcal{F}$ has first variation $\delta\mathcal{F}/\delta\rho$, the formal Otto calculus gives

$$|\partial\mathcal{F}|^2(\rho) = \|\nabla_W \mathcal{F}(\rho)\|_{T_\rho \mathcal{P}_2}^2 = \int_M \left| \nabla \frac{\delta\mathcal{F}}{\delta\rho}(\rho)(x) \right|^2 d\rho(x). \quad (2)$$

Constants in $\delta\mathcal{F}/\delta\rho$ do not matter, because only its gradient appears.

The Wasserstein gradient flow of $\mathcal{F}$ is formally

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta\mathcal{F}}{\delta\rho}(\rho_t) \right). \quad (3)$$

For smooth solutions, and more generally for curves of maximal slope when the slope is a strong upper gradient, one has the energy dissipation identity

$$\frac{d}{dt} \mathcal{F}(\rho_t) = -|\partial\mathcal{F}|^2(\rho_t) = -\|\dot{\rho}_t\|_{T_{\rho_t} \mathcal{P}_2}^2 \quad \text{for a.e. } t > 0. \quad (4)$$

The arguments below use only (4) and the functional inequalities stated explicitly.

2 Wasserstein–PL implies energy and Wasserstein-distance convergence

Definition 2.1 (Wasserstein Polyak–Łojasiewicz inequality). Let $\kappa > 0$. We say that $\mathcal{F}$ satisfies the W_2 -PL inequality with constant κ if

$$|\partial\mathcal{F}|^2(\rho) \geq 2\kappa(\mathcal{F}(\rho) - \mathcal{F}_\star) \quad \text{for every } \rho \in \text{Dom } \mathcal{F}. \quad (5)$$

Equivalently, in the smooth Otto notation,

$$\int_M \left| \nabla \frac{\delta\mathcal{F}}{\delta\rho}(\rho) \right|^2 d\rho \geq 2\kappa(\mathcal{F}(\rho) - \mathcal{F}_\star). \quad (6)$$

The factor 2 is a convention. With this convention, the logarithmic Sobolev inequality $2\kappa \text{Ent}_\pi(\rho) \leq I_\pi(\rho)$ is a W_2 -PL inequality with constant κ .

Theorem 2.2 (Energy, finite length, and Wasserstein-distance convergence). *Assume that $\mathcal{F}_\star > -\infty$ and that $(\rho_t)_{t \geq 0}$ is a global Wasserstein gradient flow satisfying the full energy dissipation identity*

$$\frac{d}{dt} \mathcal{F}(\rho_t) = -|\dot{\rho}_t|^2 = -|\partial\mathcal{F}|^2(\rho_t) \quad \text{for a.e. } t > 0. \quad (7)$$

Assume also that $\mathcal{F}$ satisfies the W_2 -PL inequality (5). Put

$$E(t) := \mathcal{F}(\rho_t) - \mathcal{F}_\star.$$

Then, for $0 \leq s \leq t$,

$$E(t) \leq e^{-2\kappa(t-s)} E(s), \quad (8)$$

and the length of the flow segment satisfies

$$\text{Length}_{W_2}((\rho_r)_{r \in [s,t]}) \leq \sqrt{\frac{2}{\kappa}} (\sqrt{E(s)} - \sqrt{E(t)}). \quad (9)$$

In particular,

$$W_2(\rho_s, \rho_t) \leq \sqrt{\frac{2}{\kappa}} (\sqrt{E(s)} - \sqrt{E(t)}). \quad (10)$$

If, in addition, $\mathcal{P}_2(M)$ is complete along the considered curve and $\mathcal{F}$ is lower semicontinuous with respect to W_2 , then ρ_t converges in W_2 to a minimizer $\rho_\infty \in \text{Argmin } \mathcal{F}$ and

$$W_2(\rho_t, \rho_\infty) \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{F}(\rho_t) - \mathcal{F}_\star} \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{F}(\rho_0) - \mathcal{F}_\star} e^{-\kappa t}. \quad (11)$$

Consequently,

$$\text{dist}_{W_2}(\rho_t, \text{Argmin } \mathcal{F}) \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{F}(\rho_0) - \mathcal{F}_\star} e^{-\kappa t}. \quad (12)$$

Proof. The energy estimate is the usual Gronwall argument: by (7) and (5),

$$E'(t) = -|\partial\mathcal{F}|^2(\rho_t) \leq -2\kappa E(t) \quad \text{for a.e. } t.$$

This gives (8).

The distance estimate uses one extra observation: PL bounds the inverse speed along the dissipating curve. On the set where $E(r) > 0$, (5) gives

$$|\partial\mathcal{F}|(\rho_r) \geq \sqrt{2\kappa E(r)}.$$

Using (7),

$$\begin{aligned} \text{Length}_{W_2}((\rho_r)_{r \in [s,t]}) &= \int_s^t |\dot{\rho}_r| \, dr = \int_s^t |\partial\mathcal{F}|(\rho_r) \, dr \\ &= \int_s^t \frac{|\partial\mathcal{F}|^2(\rho_r)}{|\partial\mathcal{F}|(\rho_r)} \, dr \\ &\leq \frac{1}{\sqrt{2\kappa}} \int_s^t \frac{|\partial\mathcal{F}|^2(\rho_r)}{\sqrt{E(r)}} \, dr \\ &= \frac{1}{\sqrt{2\kappa}} \int_s^t \frac{-E'(r)}{\sqrt{E(r)}} \, dr \\ &= \sqrt{\frac{2}{\kappa}} (\sqrt{E(s)} - \sqrt{E(t)}). \end{aligned}$$

If E vanishes on a subinterval, then ρ_r is stationary there by (7); the same estimate follows by applying the above argument on the part where $E > 0$. Since Wasserstein distance is bounded by metric length, (10) follows.

Now let $t \rightarrow \infty$. From (8), $E(t) \rightarrow 0$. From (10), the tail of the curve is Cauchy, so completeness gives a limit ρ_∞ . Lower semicontinuity yields

$$\mathcal{F}(\rho_\infty) \leq \liminf_{t \rightarrow \infty} \mathcal{F}(\rho_t) = \mathcal{F}_\star,$$

and hence $\rho_\infty \in \text{Argmin } \mathcal{F}$. Sending the endpoint of (10) to infinity with $s = t$ fixed gives the first inequality in (11); combining it with (8) gives the second. Finally, (12) follows because ρ_∞ belongs to $\text{Argmin } \mathcal{F}$. $\square$

Corollary 2.3 (Quadratic-growth/error-bound consequence). *Assume that the hypotheses of Theorem 2.2 hold for the gradient flow starting from every $\rho \in \text{Dom } \mathcal{F}$ with finite energy gap. Then the W_2 -PL inequality implies the static error bound*

$$\frac{\kappa}{2} \text{dist}_{W_2}^2(\rho, \text{Argmin } \mathcal{F}) \leq \mathcal{F}(\rho) - \mathcal{F}_*. \quad (13)$$

Equivalently,

$$\text{dist}_{W_2}(\rho, \text{Argmin } \mathcal{F}) \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{F}(\rho) - \mathcal{F}_*}. \quad (14)$$

Proof. Start the gradient flow at $\rho_0 = \rho$ and apply (12) at time $t = 0$. $\square$

Remark 2.4 (What the distance result says, and what it does not say). The estimate (12) answers the natural question: under the standard gradient-flow identity, Wasserstein–PL does give a linear rate in Wasserstein distance to the minimizer selected by the flow. The exponent is κ , while the energy exponent is 2κ , as expected after taking a square root.

This should not be confused with the stronger contraction estimate provided by positive Wasserstein geodesic convexity. If $\mathcal{F}$ is λ -geodesically convex and admits a minimizer ρ_* , the EVI theory typically gives

$$W_2(\rho_t, \rho_*) \leq e^{-\lambda t} W_2(\rho_0, \rho_*),$$

and even contraction between two solutions. Wasserstein–PL alone gives convergence to some minimizer and a distance-to-the-set estimate; it does not encode curvature along arbitrary geodesics and does not imply EVI contractivity.

Remark 2.5 (Nonunique minimizers). The estimate concerns $\text{dist}_{W_2}(\rho_t, \text{Argmin } \mathcal{F})$, or equivalently the distance to the limit minimizer ρ_∞ selected by the flow. If the minimizer is not unique, Wasserstein–PL does not imply convergence to a prescribed minimizer. The finite-dimensional model $f(x, y) = x^2$ is a useful analogy: it satisfies PL, the gradient flow sends (x_0, y_0) to $(0, y_0)$, and the distance to the solution set decays, but the distance to $(0, 0)$ need not decay.

3 Positive Wasserstein geodesic convexity implies Wasserstein–PL

Definition 3.1 (Wasserstein strong convexity). Let $\lambda > 0$. The functional $\mathcal{F}$ is λ -geodesically convex, or λ -strongly convex in the Wasserstein sense, if for every constant-speed W_2 -geodesic $(\rho_s)_{s \in [0,1]}$,

$$\mathcal{F}(\rho_s) \leq (1-s)\mathcal{F}(\rho_0) + s\mathcal{F}(\rho_1) - \frac{\lambda}{2}s(1-s)W_2^2(\rho_0, \rho_1). \quad (15)$$

Proposition 3.2 (Strong convexity $\Rightarrow$ PL). *Assume that $\mathcal{F}$ is λ -geodesically convex with $\lambda > 0$, that $\mathcal{F}_*$ is attained at some $\rho_* \in \mathcal{P}_2(M)$, and that the slope (1) is finite at ρ . Then*

$$|\partial\mathcal{F}|^2(\rho) \geq 2\lambda(\mathcal{F}(\rho) - \mathcal{F}_*). \quad (16)$$

Thus λ -Wasserstein strong convexity implies the Wasserstein–PL inequality with the same constant λ in the convention (5).

Proof. Fix ρ and a minimizer ρ_* . Put

$$g := \mathcal{F}(\rho) - \mathcal{F}_*, \quad r := W_2(\rho, \rho_*), \quad a := |\partial\mathcal{F}|(\rho).$$

If $r = 0$ there is nothing to prove. Otherwise let $(\rho_s)_{s \in [0,1]}$ be a constant-speed geodesic from ρ to ρ_* . By (15),

$$\mathcal{F}(\rho_s) \leq (1-s)\mathcal{F}(\rho) + s\mathcal{F}_* - \frac{\lambda}{2}s(1-s)r^2.$$

Hence

$$\mathcal{F}(\rho) - \mathcal{F}(\rho_s) \geq sg + \frac{\lambda}{2}s(1-s)r^2.$$

Dividing by $W_2(\rho, \rho_s) = sr$ and letting $s \downarrow 0$ in the definition of the slope yields

$$a \geq \frac{g}{r} + \frac{\lambda}{2}r. \quad (17)$$

Equivalently,

$$g \leq ar - \frac{\lambda}{2}r^2.$$

Maximizing the right-hand side over $r \geq 0$ gives

$$g \leq \frac{a^2}{2\lambda},$$

which is exactly (16). $\square$

Remark 3.3. This proof is the metric version of the elementary Euclidean argument: if f is λ -strongly convex and $x_\star$ is a minimizer, then

$$f(x) - f(x_\star) \leq \langle \nabla f(x), x - x_\star \rangle - \frac{\lambda}{2}|x - x_\star|^2 \leq \frac{1}{2\lambda}|\nabla f(x)|^2.$$

The Wasserstein proof replaces $|x - x_\star|$ by $W_2(\rho, \rho_\star)$ and the Euclidean gradient norm by the metric slope.

Remark 3.4 (Strong convexity also gives static distance control). If $\mathcal{F}$ is λ -geodesically convex with $\lambda > 0$, then applying (15) along a geodesic from a minimizer to ρ also gives

$$\mathcal{F}(\rho) - \mathcal{F}_\star \geq \frac{\lambda}{2} \text{dist}_{W_2}^2(\rho, \text{Argmin } \mathcal{F}).$$

Thus strong displacement convexity gives both PL and a static Wasserstein quadratic-growth bound. The PL theorem above shows that, for a genuine gradient flow, the distance rate can still be recovered even when geodesic convexity is absent.

4 The converse is false

The PL inequality is first-order: it compares slope to height above the minimum. Strong convexity is second-order and asks for curvature along every geodesic. These are different requirements. The following example gives a genuine Wasserstein functional satisfying W_2 -PL while failing positive Wasserstein geodesic convexity.

Example 4.1 (A nonconvex potential energy satisfying Wasserstein-PL). Work on $\mathcal{P}_2(\mathbb{R})$ and set

$$V(x) := x^2 + 2 \sin^2 x, \quad \mathcal{F}(\rho) := \int_{\mathbb{R}} V(x) d\rho(x). \quad (18)$$

Then $\mathcal{F}_\star = 0$, attained uniquely at δ_0 . The first variation is $\delta\mathcal{F}/\delta\rho = V$, so

$$|\partial\mathcal{F}|^2(\rho) = \int_{\mathbb{R}} |V'(x)|^2 d\rho(x) \quad (19)$$

for every ρ for which the right-hand side is finite. Now

$$V'(x) = 2x + 2 \sin(2x), \quad V''(x) = 2 + 4 \cos(2x).$$

We first prove a pointwise PL inequality. Since V is even and V' is odd, it is enough to consider $r = |x| \geq 0$.

If $0 \leq r \leq \pi/2$, then $\sin(2r) \geq 0$, hence

$$|V'(x)| \geq 2r, \quad V(x) \leq r^2 + 2r^2 = 3r^2,$$

and therefore

$$|V'(x)|^2 \geq \frac{4}{3}V(x). \quad (20)$$

If $r \geq \pi/2$, then $2r - 2 > 0$ and

$$|V'(x)| \geq 2r - 2, \quad V(x) \leq r^2 + 2.$$

The function $r \mapsto (2r - 2)^2/(r^2 + 2)$ is increasing for $r > 1$, so for $r \geq \pi/2$,

$$|V'(x)|^2 \geq \frac{(\pi - 2)^2}{\pi^2/4 + 2}V(x). \quad (21)$$

Combining (20) and (21),

$$|V'(x)|^2 \geq cV(x) \quad \text{for all } x \in \mathbb{R}, \quad c := \min \left\{ \frac{4}{3}, \frac{(\pi - 2)^2}{\pi^2/4 + 2} \right\} > 0. \quad (22)$$

Integrating (22) against ρ gives

$$|\partial \mathcal{F}|^2(\rho) = \int |V'|^2 d\rho \geq c \int V d\rho = c(\mathcal{F}(\rho) - \mathcal{F}_\star).$$

Thus $\mathcal{F}$ satisfies the Wasserstein–PL inequality (5) with constant

$$\kappa = \frac{c}{2} > 0. \quad (23)$$

However, $\mathcal{F}$ is not λ -geodesically convex for any $\lambda > 0$. Indeed, if it were, then restricting (15) to Dirac masses would give, for all $x, y \in \mathbb{R}$ and $s \in [0, 1]$,

$$V((1 - s)x + sy) \leq (1 - s)V(x) + sV(y) - \frac{\lambda}{2}s(1 - s)|x - y|^2,$$

which is exactly λ -strong convexity of the scalar potential V . But

$$V''(\pi/2) = 2 + 4\cos(\pi) = -2 < 0.$$

So V is not even convex, and positive Wasserstein geodesic convexity of $\mathcal{F}$ is impossible.

Remark 4.2. The example is not merely a finite-dimensional counterexample disguised as a Wasserstein one. The PL estimate holds after integration against every probability measure ρ , while failure of geodesic convexity is detected along the Wasserstein geodesics between Dirac masses. Thus the implication

$$\text{Wasserstein–PL} \implies \text{positive Wasserstein geodesic convexity}$$

fails on the full Wasserstein space.

5 A dictionary: functional inequalities as Wasserstein–PL inequalities

This section explains how many entropy–entropy production inequalities are exactly Wasserstein–PL inequalities. The guiding principle is simple:

$$\boxed{\text{energy gap } \mathcal{F}(\rho) - \mathcal{F}_\star \quad \text{versus} \quad \text{Wasserstein slope squared } |\partial\mathcal{F}|^2(\rho)}$$

A functional inequality of the form

$$2\kappa(\mathcal{F}(\rho) - \mathcal{F}_\star) \leq |\partial\mathcal{F}|^2(\rho)$$

is a Wasserstein–PL inequality by definition.

5.1 Relative entropy and logarithmic Sobolev inequality

Let π be a probability measure on $\mathbb{R}^d$ or on a Riemannian manifold, with smooth positive density, and write $\rho = h\pi$. Consider the relative entropy

$$\mathcal{F}(\rho) = \text{Ent}_\pi(\rho) := \int h \log h \, d\pi. \quad (24)$$

The first variation is $\log h$ up to an additive constant. Therefore

$$|\partial\mathcal{F}|^2(\rho) = \int |\nabla \log h|^2 h \, d\pi =: I_\pi(\rho), \quad (25)$$

the relative Fisher information. The W_2 -PL inequality for $\mathcal{F}$ is

$$2\kappa \text{Ent}_\pi(\rho) \leq I_\pi(\rho), \quad (26)$$

which is exactly the logarithmic Sobolev inequality with constant κ .

The gradient flow is

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \log \frac{\rho_t}{\pi} \right). \quad (27)$$

If $\pi = Z^{-1}e^{-V} dx$, this becomes the Fokker–Planck equation

$$\partial_t \rho_t = \Delta \rho_t + \nabla \cdot (\rho_t \nabla V). \quad (28)$$

Under (26),

$$\text{Ent}_\pi(\rho_t) \leq e^{-2\kappa t} \text{Ent}_\pi(\rho_0).$$

5.2 Relation with Poincare-type inequalities

The logarithmic Sobolev inequality is best viewed as a nonlinear, entropy-level strengthening of the Poincare spectral-gap inequality. Let

$$\pi = Z^{-1}e^{-V} dx, \quad L = \Delta - \nabla V \cdot \nabla, \quad \mathcal{E}(g, g) := \int |\nabla g|^2 d\pi,$$

so that L is symmetric in $L^2(\pi)$ and $\mathcal{E}$ is the associated Dirichlet form. The Poincare inequality with constant $\lambda > 0$ is

$$\lambda \text{Var}_\pi(g) \leq \mathcal{E}(g, g) = \int |\nabla g|^2 d\pi, \quad \int g \, d\pi = 0. \quad (29)$$

By contrast, the logarithmic Sobolev inequality, equivalently the Wasserstein–PL inequality for entropy, is

$$2\kappa \operatorname{Ent}_\pi(h) \leq I_\pi(h) := \int |\nabla \log h|^2 h \, d\pi, \quad h \geq 0, \quad \int h \, d\pi = 1. \quad (30)$$

Both inequalities have the same entropy–dissipation form along the reversible Markov semigroup $P_t = e^{tL}$. If $h_t = P_t h$, then

$$-\frac{d}{dt} \operatorname{Ent}_\pi(h_t) = I_\pi(h_t), \quad (31)$$

while

$$-\frac{d}{dt} \frac{1}{2} \int (g_t - \bar{g})^2 \, d\pi = \mathcal{E}(g_t, g_t), \quad g_t = P_t g, \quad (32)$$

where $\bar{g} = \int g \, d\pi$. Thus log-Sobolev gives exponential decay of entropy, whereas Poincare gives exponential decay of variance:

$$\operatorname{Ent}_\pi(P_t h) \leq e^{-2\kappa t} \operatorname{Ent}_\pi(h), \quad \operatorname{Var}_\pi(P_t g) \leq e^{-2\lambda t} \operatorname{Var}_\pi(g).$$

The precise local relation is obtained by linearizing (30) around equilibrium. Let $h_\varepsilon = 1 + \varepsilon g$, with $\int g \, d\pi = 0$ and ε small enough that $h_\varepsilon > 0$. Then

$$\operatorname{Ent}_\pi(h_\varepsilon) = \frac{\varepsilon^2}{2} \int g^2 \, d\pi + O(\varepsilon^3), \quad (33)$$

$$I_\pi(h_\varepsilon) = \varepsilon^2 \int |\nabla g|^2 \, d\pi + O(\varepsilon^3). \quad (34)$$

Inserting these expansions into (30) and sending $\varepsilon \rightarrow 0$ gives (29) with $\lambda \geq \kappa$. Equivalently, for the optimal constants,

$$\kappa_{\text{LSI}} \leq \lambda_{\text{PI}}. \quad (35)$$

This is the sense in which Poincare is the linearization of logarithmic Sobolev, and therefore the linearization of the Wasserstein–PL inequality for relative entropy.

The converse implication is false. Poincare controls only the quadratic, near-equilibrium part of the dynamics. Log-Sobolev controls all positive densities through the nonlinear entropy and Fisher information. A typical class of counterexamples is given by probability measures with exponential tails, such as the one-dimensional Laplace law proportional to $e^{-|x|} dx$: such measures can have a positive Poincare constant, but they do not satisfy the Gross logarithmic Sobolev inequality because log-Sobolev implies Gaussian-type concentration, while exponential tails are too heavy.

There is also a useful hierarchy. Under a positive Bakry–Emery curvature bound, or in Wasserstein language under positive displacement convexity of the entropy, one obtains both inequalities with the same curvature constant. In the notation above, the schematic implication is

$$\begin{aligned} &\text{positive curvature / Wasserstein strong convexity} \\ \implies &\quad \text{LSI} = \text{entropy Wasserstein-PL} \implies \quad \text{Poincare spectral gap.} \end{aligned} \quad (36)$$

In general none of the reverse implications is valid.

Finally, the standard Poincare inequality is not literally the Wasserstein–PL inequality for the chi-square functional. If

$$\mathcal{F}(\rho) = \frac{1}{2} \int (h - 1)^2 \, d\pi, \quad h = \frac{d\rho}{d\pi},$$

then its W_2 slope is

$$|\partial \mathcal{F}|^2(\rho) = \int h |\nabla h|^2 \, d\pi,$$

so the corresponding Wasserstein–PL inequality is the weighted estimate

$$\kappa \int (h - 1)^2 d\pi \leq \int h |\nabla h|^2 d\pi.$$

The usual Poincaré inequality has no mobility factor h on the right. Hence it is more accurately a PL/entropy-production inequality for the linear Markov semigroup and its quadratic entropy, whereas log-Sobolev is the Wasserstein–PL inequality for the entropy functional.

Example 5.1 (Gaussian case). For the standard Gaussian measure γ on $\mathbb{R}^d$,

$$2 \operatorname{Ent}_\gamma(\rho) \leq I_\gamma(\rho).$$

Thus the entropy is a W_2 -PL functional with $\kappa = 1$. The Ornstein–Uhlenbeck/Fokker–Planck flow has entropy decay e^{-2t} , and Theorem 2.2 also gives

$$W_2(\rho_t, \gamma) \leq \sqrt{2 \operatorname{Ent}_\gamma(\rho_0)} e^{-t}.$$

5.3 Talagrand and quadratic-growth inequalities as distance consequences

The distance estimate in Theorem 2.2 is not itself the Wasserstein–PL inequality; it is a consequence of Wasserstein–PL plus the energy dissipation identity. In the entropy case, it has the same shape as a Talagrand T_2 inequality. Indeed, if $\operatorname{Argmin} \operatorname{Ent}_\pi = \{\pi\}$ and the entropy gradient flow is global, then (14) gives

$$W_2^2(\rho, \pi) \leq \frac{2}{\kappa} \operatorname{Ent}_\pi(\rho), \quad (37)$$

whenever the flow can be started from ρ . For entropy, the classical Otto–Villani implication “log-Sobolev implies Talagrand” gives the same type of static estimate.

Thus there are three related but logically distinct inequalities:

$$\begin{aligned} \text{Wasserstein–PL / LSI: } & 2\kappa \operatorname{Ent}_\pi(\rho) \leq I_\pi(\rho), \\ \text{Talagrand / transport inequality: } & W_2^2(\rho, \pi) \leq \frac{2}{\kappa} \operatorname{Ent}_\pi(\rho), \\ \text{flow distance rate: } & W_2(\rho_t, \pi) \leq \sqrt{\frac{2}{\kappa} \operatorname{Ent}_\pi(\rho_0)} e^{-\kappa t}. \end{aligned}$$

The first inequality compares entropy with entropy production; the second compares transport distance with entropy; the third is a dynamical consequence along the gradient flow.

5.4 Φ -entropy inequalities

Let $\Phi : (0, \infty) \rightarrow \mathbb{R}$ be convex, with $\Phi(1) = \Phi'(1) = 0$, and set

$$\mathcal{F}_\Phi(\rho) = \int \Phi(h) d\pi, \quad h = \frac{d\rho}{d\pi}. \quad (38)$$

The first variation is $\Phi'(h)$, hence

$$|\partial \mathcal{F}_\Phi|^2(\rho) = \int h |\nabla \Phi'(h)|^2 d\pi. \quad (39)$$

The Wasserstein–PL inequality is therefore the Φ -entropy production inequality

$$2\kappa \int \Phi(h) d\pi \leq \int h |\nabla \Phi'(h)|^2 d\pi. \quad (40)$$

This formula is a general template. The classical name of the inequality depends on the choice of Φ .

5.5 A weighted Poincare or chi-square production inequality

Take

$$\Phi(r) = \frac{1}{2}(r-1)^2.$$

Then

$$\mathcal{F}(\rho) = \frac{1}{2} \int (h-1)^2 d\pi = \frac{1}{2} \chi^2(\rho \mid \pi), \quad |\partial \mathcal{F}|^2(\rho) = \int h |\nabla h|^2 d\pi.$$

The corresponding Wasserstein–PL inequality is

$$\kappa \int (h-1)^2 d\pi \leq \int h |\nabla h|^2 d\pi. \quad (41)$$

This is a weighted Poincare-type inequality, or a chi-square entropy production inequality, with the Wasserstein mobility h on the right-hand side. It should not be confused with the standard Poincare inequality

$$\kappa \int (h-1)^2 d\pi \leq \int |\nabla h|^2 d\pi,$$

although the two are directly comparable on classes where h is bounded below or above.

The associated Wasserstein gradient flow is nonlinear:

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla h_t), \quad h_t = \frac{d\rho_t}{d\pi}. \quad (42)$$

5.6 Power entropies, Beckner-type inequalities, and porous-medium structure

For $m > 1$, define the normalized power entropy

$$\Phi_m(r) = \frac{r^m - mr + m - 1}{m - 1}. \quad (43)$$

Then $\Phi_m(1) = \Phi'_m(1) = 0$ and

$$\Phi'_m(r) = \frac{m}{m-1} (r^{m-1} - 1).$$

Using (39),

$$\begin{aligned} |\partial \mathcal{F}_{\Phi_m}|^2(\rho) &= m^2 \int h^{2m-3} |\nabla h|^2 d\pi \\ &= \frac{4m^2}{(2m-1)^2} \int \left| \nabla h^{m-1/2} \right|^2 d\pi. \end{aligned} \quad (44)$$

Thus the W_2 -PL inequality is

$$2\kappa \int \frac{h^m - mh + m - 1}{m - 1} d\pi \leq \frac{4m^2}{(2m-1)^2} \int \left| \nabla h^{m-1/2} \right|^2 d\pi. \quad (45)$$

Depending on the reference measure and the range of exponents, inequalities of this type appear as Beckner, convex Sobolev, generalized Sobolev, or interpolation inequalities. In the limit $m \downarrow 1$, (45) formally recovers the logarithmic Sobolev inequality.

The free diffusion energy

$$\rho \mapsto \frac{1}{m-1} \int_{\mathbb{R}^d} \rho^m dx$$

has Wasserstein gradient flow

$$\partial_t \rho_t = \Delta(\rho_t^m),$$

the porous-medium equation. In confined or self-similar variables, the corresponding relative entropy–production inequality gives exponential convergence to the Barenblatt equilibrium and is closely related to sharp Gagliardo–Nirenberg–Sobolev inequalities.

5.7 Nonlinear Fokker–Planck and granular-media free energies

Consider a free energy on $\mathcal{P}_2(\mathbb{R}^d)$ of the form

$$\mathcal{F}(\rho) = \int \rho \log \rho \, dx + \int V(x) \, d\rho(x) + \frac{1}{2} \iint W(x-y) \, d\rho(x) \, d\rho(y). \quad (46)$$

The first variation is

$$\frac{\delta \mathcal{F}}{\delta \rho} = \log \rho + 1 + V + W * \rho,$$

so the Wasserstein slope squared is

$$|\partial \mathcal{F}|^2(\rho) = \int |\nabla \log \rho + \nabla V + \nabla W * \rho|^2 \, d\rho. \quad (47)$$

The W_2 -PL inequality is the entropy–entropy production estimate

$$2\kappa(\mathcal{F}(\rho) - \mathcal{F}_*) \leq \int |\nabla \log \rho + \nabla V + \nabla W * \rho|^2 \, d\rho. \quad (48)$$

The associated gradient flow is the aggregation–diffusion equation

$$\partial_t \rho_t = \nabla \cdot (\rho_t \nabla (\log \rho_t + V + W * \rho_t)). \quad (49)$$

When the energy is positively displacement convex, (48) follows from Proposition 3.2. In some nonconvex settings, an inequality of the form (48) may still hold and then gives exponential decay of the free energy without geodesic convexity.

5.8 HWI as a route to PL for entropy

For entropy relative to a measure π satisfying a positive curvature condition, the HWI inequality has the schematic form

$$\text{Ent}_\pi(\rho) \leq W_2(\rho, \pi) \sqrt{I_\pi(\rho)} - \frac{K}{2} W_2^2(\rho, \pi), \quad K > 0. \quad (50)$$

Optimizing the right-hand side over the distance variable gives

$$\text{Ent}_\pi(\rho) \leq \frac{1}{2K} I_\pi(\rho), \quad (51)$$

that is, the logarithmic Sobolev inequality and hence the Wasserstein–PL inequality for entropy. Once this PL inequality is available, Theorem 2.2 gives the corresponding flow estimate

$$W_2(\rho_t, \pi) \leq \sqrt{\frac{2}{K} \text{Ent}_\pi(\rho_0)} e^{-Kt}$$

for the entropy gradient flow. Thus HWI is not itself the PL inequality, but it is a standard mechanism producing PL; the PL inequality plus the gradient-flow identity then gives the distance rate.

5.9 Summary table

Energy $\mathcal{F}(\rho)$	Slope squared $ \partial\mathcal{F} ^2$	W_2 -PL inequality / usual name
$\text{Ent}_\pi(\rho) = \int h \log h \, d\pi$	$I_\pi(\rho) = \int \nabla \log h ^2 h \, d\pi$	$2\kappa \text{Ent}_\pi \leq I_\pi$; logarithmic Sobolev inequality.
$\frac{1}{2}\chi^2(\rho \mid \pi)$	$\int h \nabla h ^2 \, d\pi$	Weighted Poincare / chi-square entropy production inequality.
$\int \Phi(h) \, d\pi$	$\int h \nabla \Phi'(h) ^2 \, d\pi$	Φ -entropy production inequality.
Power entropy $\int \Phi_m(h) \, d\pi$	$\frac{4m^2}{(2m-1)^2} \int \nabla h^{m-1/2} ^2 \, d\pi$	Beckner, convex Sobolev, generalized Sobolev, or interpolation inequality.
Confined porous-medium entropy	$\int \rho \nabla \psi_\rho ^2 \, dx,$ $\psi_\rho = \frac{m}{m-1} \rho^{m-1} + V$	Entropy–entropy production inequality; often equivalent to Gagliardo–Nirenberg–Sobolev inequalities in self-similar variables.
Granular-media free energy (46)	Dissipation (47)	Entropy–entropy production inequality for aggregation–diffusion.

6 Takeaways

- (i) A Wasserstein–PL inequality is the metric-slope inequality

$$|\partial\mathcal{F}|^2 \geq 2\kappa(\mathcal{F} - \mathcal{F}_\star).$$

Together with the Wasserstein energy dissipation identity, it gives the energy rate

$$\mathcal{F}(\rho_t) - \mathcal{F}_\star \leq e^{-2\kappa t}(\mathcal{F}(\rho_0) - \mathcal{F}_\star).$$

- (ii) The same assumptions also give a distance rate. The key estimate is the finite-length bound

$$\text{Length}_{W_2}((\rho_s)_{s \geq t}) \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{F}(\rho_t) - \mathcal{F}_\star}.$$

Hence, under lower semicontinuity and completeness assumptions,

$$\text{dist}_{W_2}(\rho_t, \text{Argmin } \mathcal{F}) \leq \sqrt{\frac{2}{\kappa}} \sqrt{\mathcal{F}(\rho_0) - \mathcal{F}_\star} e^{-\kappa t}.$$

The exponent is half the energy exponent because distance is controlled by the square root of the energy gap.

- (iii) Positive Wasserstein geodesic convexity is sufficient for Wasserstein–PL and gives stronger EVI/contractivity information. Wasserstein–PL by itself gives convergence to the minimizer selected by the flow, but not curvature along arbitrary Wasserstein geodesics.
- (iv) The converse is false. The potential energy $\mathcal{F}(\rho) = \int (x^2 + 2 \sin^2 x) \, d\rho(x)$ satisfies Wasserstein–PL because the pointwise scalar PL inequality $|V'|^2 \geq cV$ holds, but it is not positively Wasserstein geodesically convex because $V''(\pi/2) < 0$.
- (v) Many standard functional inequalities are simply Wasserstein–PL inequalities written for specific choices of $\mathcal{F}$. The most important example is the logarithmic Sobolev inequality, which is the Wasserstein–PL inequality for relative entropy. Talagrand-type inequalities are not PL inequalities, but they are closely related distance/error-bound consequences.
- (vi) Poincare-type inequalities are the quadratic or linearized counterparts of these entropy inequalities. Logarithmic Sobolev implies Poincare by linearization around the equilibrium

density $h \equiv 1$, but Poincare does not imply logarithmic Sobolev in general. For the entropy functional one has the hierarchy

$$\begin{aligned} &\text{positive Wasserstein curvature} \\ &\implies \text{log-Sobolev} / \text{Wasserstein-PL} \implies \text{Poincare spectral gap.} \end{aligned}$$

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